

Role Support in Knowledge-Graph Error Ranking: Predictor Regimes and Evaluation Policies

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Missing support has no fixed error meaning

RESEARCH QUESTION

When do training-role counts improve error ranking for a fixed knowledge-graph predictor, and how does the answer change across predictor and decision policies?

0.689 → 0.364

raw-score AUROC across FB15k-237 regimes

12 / 12

ensemble/head AURC and Brier gains from counts

2

static graphs; exploratory reuse

Outcome and information

Predict a tail for each (head, relation) query. Error means the prediction is absent from the recorded held-out target set. This is recorded-answer recovery, not factual falsehood. Counts use training data; endpoint features use the predicted tail, never a hidden answer.

Positive feature result

Continuous role counts improve AURC (error across answer rates) and Brier (error-probability scoring) beyond confidence, frequency, and ensemble disagreement in all 12 fixed-ensemble/error-head comparisons. This is conditional on six fixed ensembles and two error-head families, not independent graph replication or uniform operating-point dominance.

Policy matters

Restricting candidates to training-supported tails raises DistMult accuracy on FB15k-237 but lowers it on WN18RR. A development-fixed acceptance threshold also changes the test answer fraction, so a lower error ratio can accompany fewer answers.

Raw-score failure

On FB15k-237, original versus continued ComplEx raises mean observed-target accuracy from 10.05% to 44.35%, while the positive missing-role score's error AUROC reverses from 0.6893 to 0.3639. Multiple training changes occur together; accuracy is not isolated as the cause.

Representation matters

Zero-count bits are deterministic functions of counts. They add no conditional information, but can change a regularized fit. Counts logistic regression reaches the 85%-answer-rate hindsight floor in all 12 original runs and none of the six continued-training runs.

EVIDENCE AT A GLANCE

Role-restricted candidates	FB15k-237	WN18RR
Accuracy before	10.56%	38.16%
Accuracy after	25.79%	28.05%
Targets unreachable	6.63%	27.67%

TAKEAWAY

Learn from role counts when they help the specified predictor and decision policy. Do not assign training absence a universal uncertainty or factual-truth interpretation.

INTERPRETATION & LIMITS

Custom recorded-target populations; two inspected static graphs; no temporal transfer or independent confirmation. Seeds and ensembles are not graph replications. Training changes do not identify a causal mechanism. Historical results remain unreproduced. No conformal guarantee, posterior uncertainty decomposition, or calibrated factual-truth claim.